

Cover letter — Geodata and AI (Elsevier)

Dear Editors,

Please consider the enclosed manuscript, "A national inventory of well lithology and hydraulics for the conterminous United States, with a vision-language method for reading scanned driller's logs," for publication as a Research Article in Geodata and AI.

The work sits squarely at the intersection of the journal's two core interests — AI methods for the geosciences and the compilation and sharing of open geo-databases. It contributes both:

(1) A geo-database. We assemble and openly release a harmonized national inventory of more than 23.9 million depth-resolved lithology intervals across 3.4 million water wells (17 states), plus structured hydraulics for about 2.8 million wells, drawn entirely from public state-agency endpoints with no privileged access. To our knowledge this is the first openly-licensed, depth-resolved subsurface-lithology layer assembled at national scale from primary driller's-log sources; it fills a gap that continental groundwater models currently approximate with coarse, surface, map-derived geology.

(2) A GeoAI method. We introduce a reproducible, cost-bounded pipeline that uses vision-language models to recover structured lithology directly from scanned driller's-log images — no OCR stage, no per-form templates, no per-state training. It is validated against a state agency's own hand-transcriptions (96.0% coarse/fine accuracy, Cohen's kappa = 0.92) and metered in provider-independent token units (about 1,000 tokens per log), so the economics are reproducible by any group. We also report an honest, systematically-probed national availability census of what each state does and does not expose.

We believe this is a strong fit for Geodata and AI specifically because the journal recognizes geo-databases as a first-class contribution and foregrounds AI for geoscientific discovery — the two halves of this paper. The frozen dataset and the essential method code are deposited in a public repository (Zenodo) with a citable DOI (activated on acceptance) and cited in the manuscript; a preprint policy-compliant arrangement.

The work is original, not under consideration elsewhere, and has not been previously published. A declaration of competing interests (the author maintains the SWATGenX platform that hosts an optional interactive viewer), a funding statement (none), a CRediT statement, and a declaration of generative-AI use in manuscript preparation (distinct from the vision-language models used as the research method) are all included. We would be glad to suggest reviewers with expertise in groundwater data systems and AI-assisted document extraction upon request.

This manuscript is submitted before 31 August 2026, and we respectfully note the journal's APC (article processing charge) waiver for submissions received by that date.

Thank you for your consideration.

Sincerely,

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